

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 22-Nov-2022

Time: Three Hours

Max. Marks: 75 Marks

Physical Pharmaceutics - II

Q.P. CODE: 5015

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Discuss the electrical properties and kinetic properties of colloids.

OR

Define Thixotropy. Explain different methods for its determination and give its application in pharmacy.

2. How do you prepare flocculated suspension? Define sedimentation volume and degree of Flocculation.

SHORT ESSAYS

7 x 5 = 35 Marks

3. What are colloids? Classify the colloids. Differentiate between different types of colloids.

OR

Explain methods of purification of colloids with the help of neat labeled diagram.

4. Discuss plastic and pseudoplastic system of flow.

OR

Differentiate flocculated and de-flocculated suspension.

5. Enumerate methods to determine the particle size. Explain any two methods to determine the particle size.
6. What are derived properties of powders? Explain any two.
7. What is specific surface area? How is it measured by air permeability method?
8. Explain the factors influencing the rate of reaction.
9. Define order of reaction. Explain the substitution method for determination of order of reaction.

SHORT ANSWERS

10 x 2 = 20 Marks

10. What is Kraft point?
11. What is Hardy Schulze Rule?
12. Define Viscosity along with its units of expressions.
13. Define Newton's law of flow with equation.
14. Define molecularity of reaction.
15. Define micro emulsion and multiple emulsions.
16. State stokes law.
17. What is particle size distribution and particle number?
18. Differentiate between half life and shelf life for a first order reaction.
19. Define pseudo zero order reaction with example.
